

Q.18 Explain any four general maintenance steps for hand tools or electronic instruments.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 What is the purpose of test equipment recalibration? Explain the procedure to recalibrate a multimeter and describe the risks of using an uncalibrated device.

Q.20 Evaluate the role of PPE as a last line of defense in hazardous work environments. How should employers and employees share responsibility for proper PPE usage and upkeep?

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Level 5, 2nd Sem / DVOC (Medical Image Tech.)

Subject : Tools, Equipment & Safety Measures - II

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 In cable termination, crimping is performed to:

- a) Test voltage
- b) Remove insulation
- c) Secure a connector to a cable
- d) Polish the cable end

Q.2 Which test instrument needs periodic recalibration to maintain measurement precision in low-voltage systems?

- a) Wire cutter
- b) Oscilloscope
- c) Screwdriver
- d) Soldering iron

Q.3 Antistatic wrist straps are used to:

- a) Prevent noise hazards
- b) Protect from fire
- c) Ground the technician and prevent ESD damage
- d) Replace power cables

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- Q.4 PPE must be stored in a:
- a) Clean, dry, and ventilated area
 - b) Toolbox with oily tools
 - c) Damp, open environment
 - d) Locked drawer without ventilation
- Q.5 Which of the following is a Don't in a cleanroom environment?
- a) Wearing gloves
 - b) Walking slowly and carefully
 - c) Following entry protocols
 - d) Using personal mobile phones

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 In electronic servicing, color-coded wires are used to aid in quick _____ of circuits.
- Q.7 General safety includes hazard awareness, while _____ safety involves individual responsibility and decision-making.
- Q.8 Personal protective gear must comply with _____ standards to ensure proper protection levels.
- Q.9 Earmuffs and earplugs are categorized under _____ PPE.

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- Q.10 Air inside a cleanroom is filtered through _____ filters to remove microscopic contaminants.

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

- Q.11 List and explain four different types of tools and equipment used in installation and servicing of electronic systems.
- Q.12 Define electronics Manufacturing Services (EMS). Highlight its role in modern electronic product assembly with suitable industry examples.
- Q.13 Enumerate five common safety precautions one must follow while handling precision tools.
- Q.14 Explain the types of PPE used in electronics manufacturing, highlighting usage scenarios for each.
- Q.15 Write a maintenance checklist for Personal Protective Equipment (PPE) used in high-voltage lab environments.
- Q.16 Write five Do's and five Don'ts in a clean Room environment.
- Q.17 Describe the process of identifying and terminating coaxial cables in electronic servicing.

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